

Decisions at the Grid's Edge: An Improved Spatial PV Expansion in Swiss Residences

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Motivation

Energiewende

Solarstrom: Schweiz nutzt zehn Prozent der geeigneten Dachflächen

Solarstrom legt weiter zu: Erstmals nutzt die Schweiz zehn Prozent der geeigneten Dachflächen. Wie viel tatsächlich nötig ist – und wo ihre Gemeinde steht, zeigt unsere Grafikübersicht.

Der Solar-Boom bringt das Stromnetz an den Anschlag

Weil im Sommer massenweise Solarstrom produziert wird, droht das Verteilnetz aus dem Gleichgewicht zu geraten. Braucht es bereits wieder einen Kurswechsel bei der Förderung von Sonnenenergie?

Why rooftop solar in Mass. is flawed, and how to fix it

It turns a climate solution into a burden for residents without solar, raising their electric bills.

Erneuerbare Energien

Solarbranche: Ausbau erneuerbarer Energien ist Bürgerwille

Leitfaden zur Energiewende

Verschrottet den Solar-Express

Die blindwütige Fixierung auf Solar-Grossanlagen in den Alpen erweist sich als Irrweg. Wenn wir genug Strom haben wollen, braucht es ein Umdenken.

Motivation

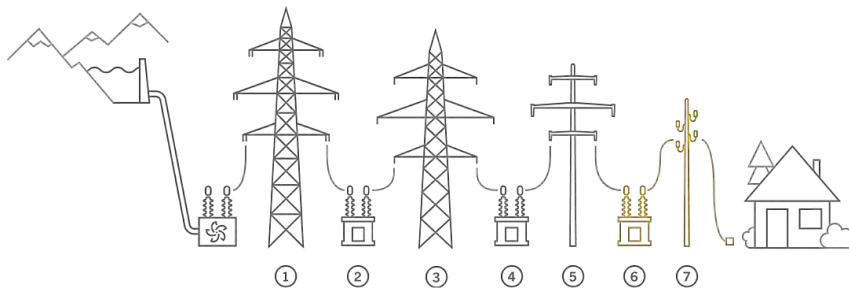
To achieve the energy transition, (residential) PV plays a vital role...

- Surpass coal in 2027 with 2'350 GW (22%) of global power production capacity (IEA, [2022](#))
- Has a potential of 67 TWh/y on existing Swiss infrastructure (AIEP, [2021](#))
- Targeted to have 37.5 GW capacity in Switzerland in 2050 ([SFOE], [2020](#), 8.17 GW in 2024 SwissSolar, [2024](#), [◀ App.Res Targets](#))

... but faces serious problems.

- Recent political opposition.
- Dislike for large scale installations (Sütterlin and Siegrist, [2017](#), Cousse et al., [2020](#))
- Grid overloading has already been observed and requires costly counter-measures (REN21, [2023](#), Tomar, [2024](#))

Grid Structure



Literature

- What drives PV adaptation (Müller and Trutnevyte, 2020)
- How to assess the PV induced problem on "hosting capacity" (Ismael et al., 2019, Mulenga et al., 2020)
- What levels of PV-penetration is problematic (Tear et al., 2019)
- Optimal placement of PV installations and energy storage for minimal medium voltage grid strain (Gupta et al., 2021)
- Level of investments needed to facility energy transition (incl. high levels of residential PV, SFOE, 2022, ◀ App.Grid Analysis).

Research Contribution - A Swiss Case Study

1 Problem quantification:

- Create highly detailed spatial model to estimate PV-induced effects in larger LV grid (incl. roof specifics + individual "decision making").

2 Economic response:

- 1st-best) Congestion pricing for externality (not feasible in practice)
- 2nd-best) Simple subsidy adjustments, focusing on grid capacity

Model Structure - 3 Agents

Demand

- Archetype profiles (single/multi family, rural / urban) scaled by living area.
- Scaling up winter months for heat pumps.



Production

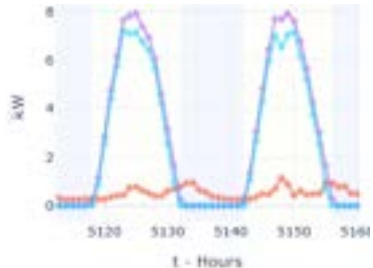
- Roof partitions and radiation yield production potential.
- Calibrated random forest assigns installation size

◀ App. RFR

Investments

- General cost assumption of PV calculator

◀ App. Cost



$$\rightarrow NPV_i = -Inv_i + \sum_t^{25} \frac{Sav_{i,t} + Feed_{i,t}}{(1+r)^t}$$

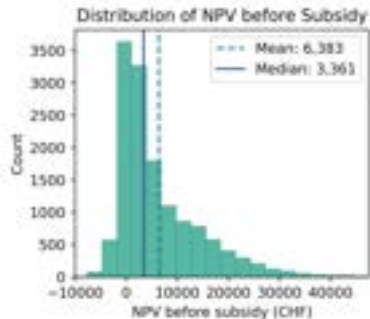
Model Structure - Allocation Process

Allocation process

- Installation "order" defined by NPV.
- Scaled, annual "construction budget" (SFOE, 2020) until 2050.

Subsidy adjustment

- Subsidy budget given to operator (transparent criteria)
- 3 types (A: roof orientation, B: grid specific, C: combination)



	$S = 0$	2000	4000	6000
$R = 0$	BU	A		
4000	B	C		
6000				
8000				

$$\rightarrow NPV_{i,X} = NPV_i + S_{i,X} - R_{i,X}$$

Data

Summary

Metric	Sample	Avail. Sample	All CH
Municipalities	28	79	2132
All Buildings	44'354	132'018	-
Residential Houses	15'435	47'900	-
Grid Nodes	334	1'113	-
Total Roof Surface (m ²)	2'322'657.0	20'788'344.3	-
Capacity Target 2050 (MW)	129.2	-	-

Data Sources

Category	Description	Ref.
Rooftop Data	Roof surface, orientation, tilt	[1]
Building Data	Building type, heating tech, existing inst. prices / tariffs	[2–5]
Grid Infrastructure	Nodes, connections + capacity	(NDA)
Other Data	Demand, costs assumptions, capacity targets, radiation	[6–9]
Computing Infrastructure	High Performance Cluster for model run	[10]

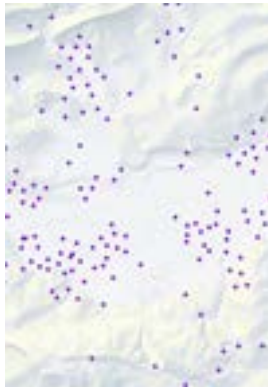
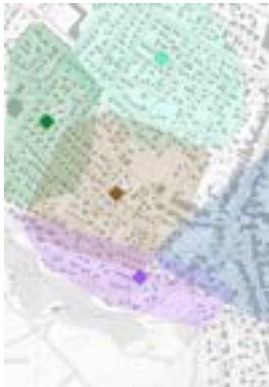
[1]: SFOE, [2023](#)

[2–5]: AIEP, [2023](#); SFSO, [2025](#); Swiss Federal Electricity Commission [ElCom], [2023](#); SFOE, [2024](#)

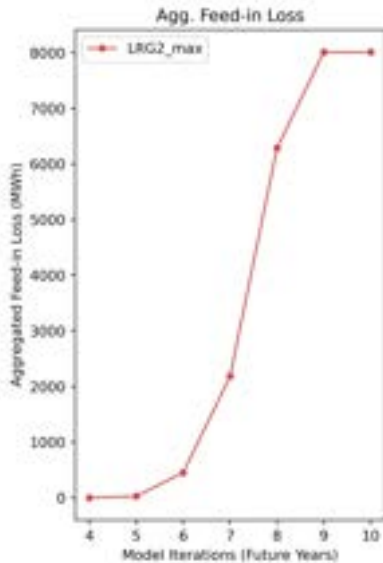
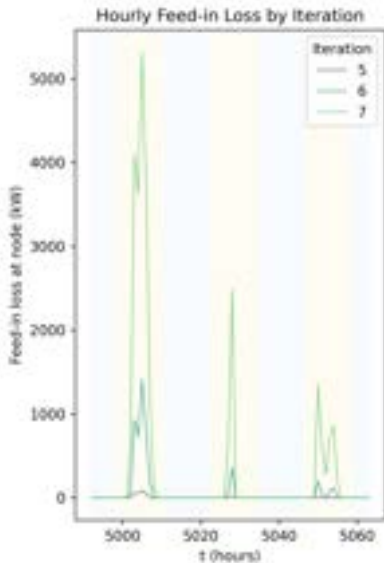
[6–9]: SFOE, [2020](#); energieschweiz, [2023](#); meteoblue, [2023](#); Rinaldi et al., [2022](#)

[10]: HPC University of Basel, [2026](#)

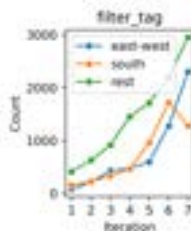
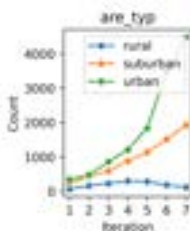
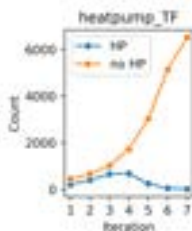
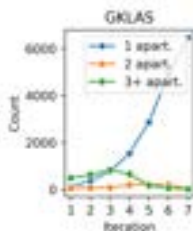
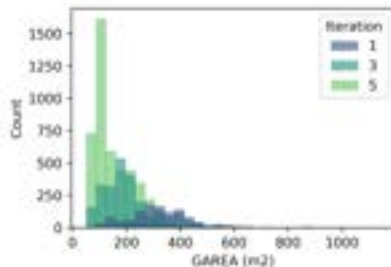
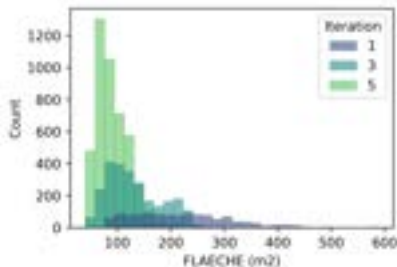
Data - Figure



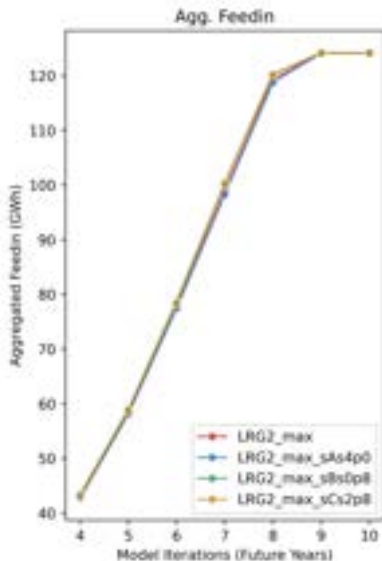
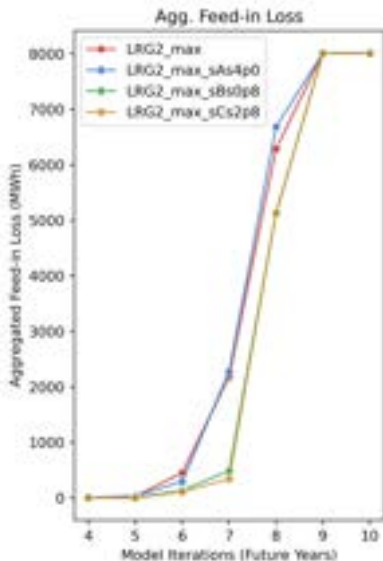
Future Grid-Related Losses



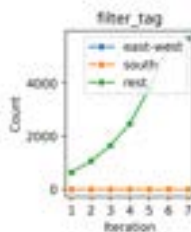
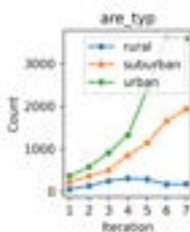
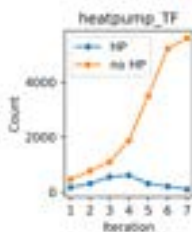
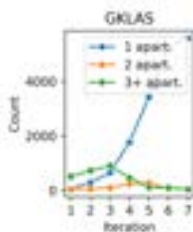
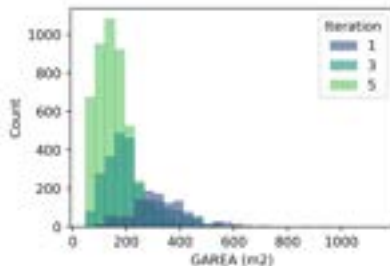
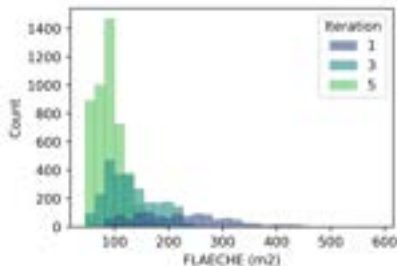
Characteristics



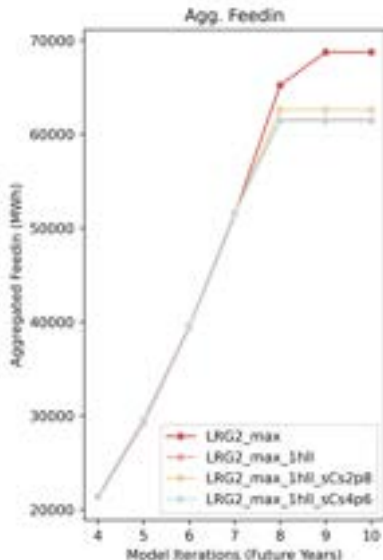
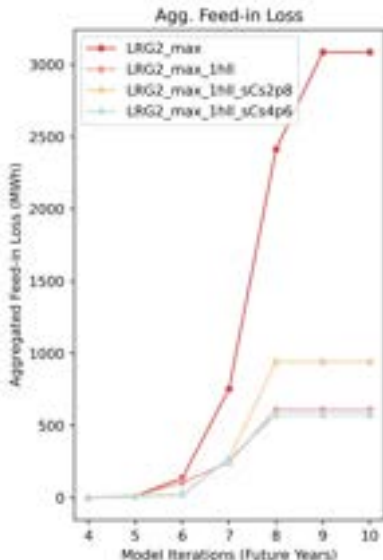
"Subsidies A,B and C"



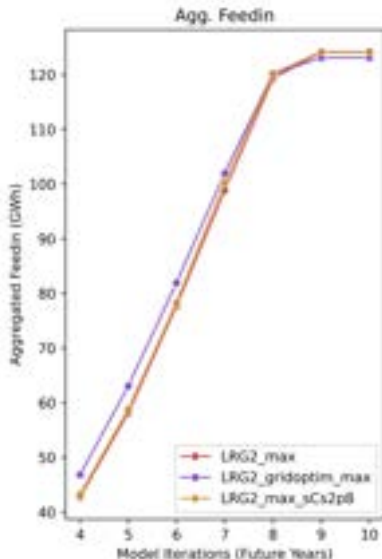
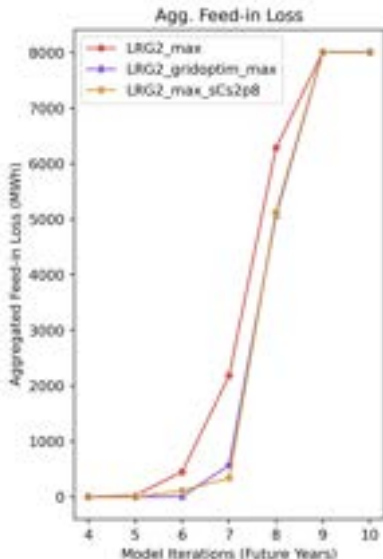
"Subsidy C"



"Cautious Grid Operator"



"Optimal Expansion" (in developing)



Conclusion

Research Contribution

1 Problem quantification:

- *Problems will arise in the next couple of years and scale up fast*

2 Economic response:

- *Simple subsidy adjustments can delay grid congestion by a few years and overall size, possibly maybe even as good as a social planners allocation.*

Outlook

- Derive "optimal installation schedule" for comparison (*in progress)
- Add battery investment and individual battery usage to the model
→ optimize household consumption
- More sophisticated pricing schemes (actual congestion pricing)
- Use RTP instead of TOU prices for electricity self consumption.

Thank you

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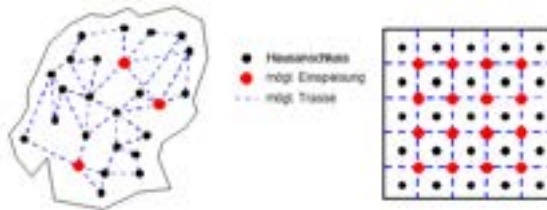
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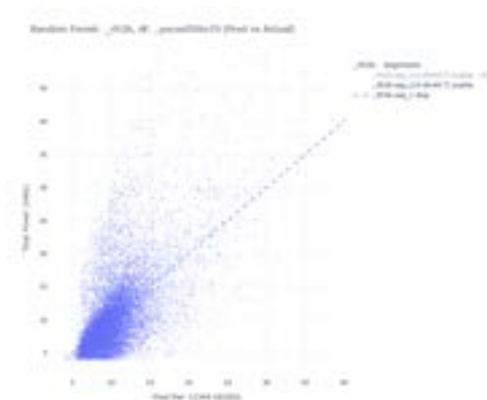
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Appendix: Model Grid Analysis EP2050+



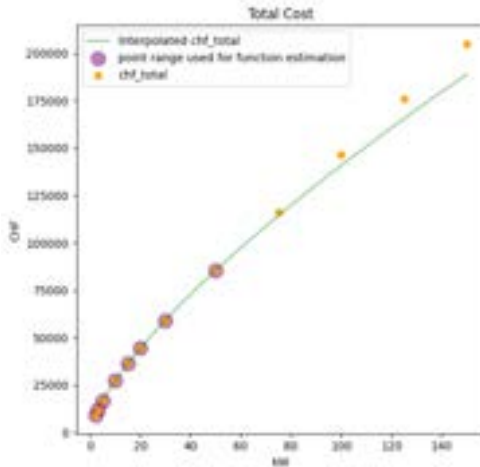
Comparison between the real (left) and homogeneous model assumed grid topology SFOE, 2022 [◀ Literature](#)

Appendix: Calibrated Random Forest



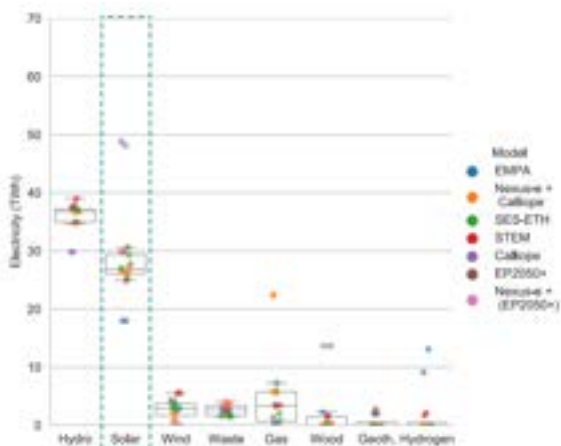
Scatterplot for actual and predicted installation sizes (incl 45 degree diagonal) of calibrated random forest regression on 61270 Swiss houses with a PV installation, between 20 to 70 percent an available roof partition used. [◀ back](#)

Appendix: Interpolated Cost Function



*Intrapolation approximation function (green, only on small scale installations)
based on cost assumptions in the technical model appendix of
energieschweiz.* [◀ back](#)

Appendix: RES Targets 2050 Switzerland

[◀ back](#)